

05

PHASE

Business Intelligence & Dashboards

Turn Data Into Insights.
Visualize. Analyze. Act.



Power BI



Tableau



Google Sheets



Excel



Connect
Data



Model
Data



Visualize
Insights



Make Better
Decisions



Drive Business
Impact

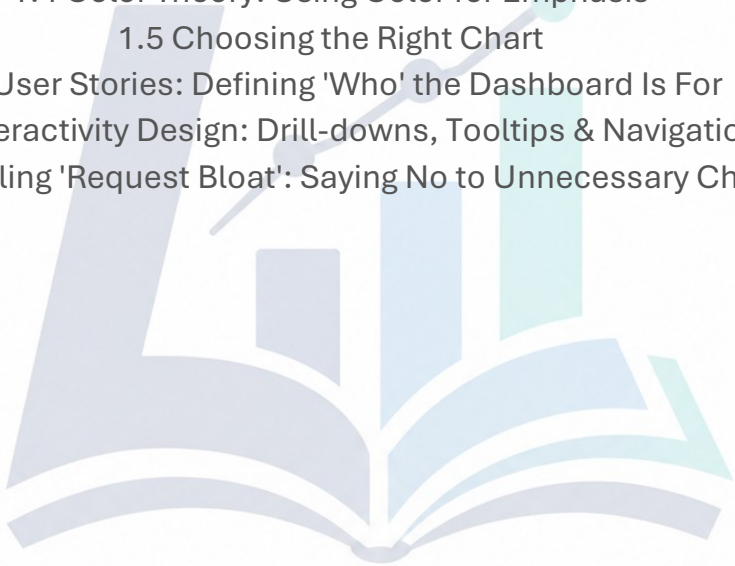


Better Insights. Better Decisions. **Better Outcomes.**

UNIT 1: Dashboard Design & UX Strategy

Topics Covered

- 1.1 Defining KPIs: Lead vs. Lag Indicators
- 1.2 The 5-Second Rule: Information Hierarchy in Layouts
- 1.3 Visual Perception: Gestalt Principles
- 1.4 Color Theory: Using Color for Emphasis
- 1.5 Choosing the Right Chart
- 1.6 User Stories: Defining 'Who' the Dashboard Is For
- 1.7 Interactivity Design: Drill-downs, Tooltips & Navigation
- 1.8 Handling 'Request Bloat': Saying No to Unnecessary Charts



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1.1 Defining KPIs: Lead vs. Lag Indicators

Definition

A Key Performance Indicator (KPI) is a measurable value that shows how effectively a business is achieving its goals. Think of KPIs as the vital signs of a business — they tell you whether things are healthy or not.

Sub-Topics

Lead Indicators

Lead indicators are forward-looking metrics. They predict future outcomes and are things you can directly influence or act on right now. They give you early warning signals before results arrive.

Examples: Number of sales calls made this week
Website visitors per day
Number of product demos scheduled

Lag Indicators

Lag indicators are backward-looking metrics. They measure what has already happened — the final outcome or result. You cannot change them; you can only learn from them.

Examples: Total revenue earned last quarter
Customer churn rate last month
Annual profit margin

Aspect	Lead vs. Lag Comparison
Direction	Lead = Future-focused Lag = Past-focused
Actionability	Lead = Directly controllable Lag = Not directly changeable
Timing	Lead = Early signal Lag = Final result
Use Case	Lead = Daily operations Lag = Performance reviews
Example (Sales)	Lead = Calls made Lag = Revenue earned

Why Important

- Without KPIs, you are flying blind — you have no idea if you are winning or losing.
- Lead indicators allow analysts to take corrective actions before it is too late.

- Lag indicators confirm whether your strategies actually worked.
- A good dashboard always shows BOTH types — prediction AND confirmation.

 Real-World Analogy

Analogy Think of driving a car. The SPEEDOMETER is a lead indicator — it tells you how fast you are going right now, and you can press the brakes immediately. Your FUEL BILL at the end of the month is a lag indicator — it tells you how much you spent, but you can't change it. Smart drivers watch both.

 Example

An e-commerce company tracks:

- Lead: Number of product page views today → predicts sales tomorrow
- Lead: Cart abandonment rate this hour → signals pricing or UX issues
- Lag: Total orders completed last week → confirms sales team performance
- Lag: Revenue vs. target for the month → shows if goals were met

 Industry Use Case

Industry	Lead & Lag KPI Application
Healthcare	Lead: Patient wait time today Lag: Monthly patient satisfaction score
Retail	Lead: Daily foot traffic in store Lag: Weekly same-store sales growth
HR	Lead: Number of interviews scheduled Lag: Employee turnover rate
Finance	Lead: Loan applications received Lag: Default rate last quarter
Marketing	Lead: Email open rate Lag: Customer acquisition cost

1.2 The 5-Second Rule: Information Hierarchy in Layouts

 Definition

The 5-Second Rule states that a dashboard user should understand the most important insight within 5 seconds of looking at it. If they can't, the layout is poorly designed. This forces analysts to prioritize what goes where on a dashboard.

Information Hierarchy

Information hierarchy means organizing content by importance — the most critical information occupies the most prominent position (top-left for most readers), and less critical details appear further down or in smaller sizes.

The F-Pattern & Z-Pattern

Research shows human eyes follow predictable patterns when scanning screens:

- F-Pattern: Eyes scan across the top, then down the left side. Used for data-heavy dashboards.
- Z-Pattern: Eyes scan top-left → top-right → bottom-left → bottom-right. Used for simpler executive dashboards.

The Three Zones of a Dashboard

Zone	Content & Priority Level
Zone 1 – Top Left	PRIMARY: Most critical KPI (e.g., Revenue Today, System Status)
Zone 2 – Top Right	SECONDARY: Supporting context (e.g., trend line, comparison)
Zone 3 – Bottom Half	TERTIARY: Detail tables, filters, drill-down charts

⚡ Why Important

- Executives make faster decisions when information is prioritized correctly.
- Poorly laid out dashboards cause 'dashboard blindness' — users stop trusting them.
- The 5-second rule prevents chart junk — unnecessary visuals that waste attention.
- Following hierarchy reduces cognitive load and prevents mental fatigue.

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🌐 Real-World Analogy

Analogy Think of a newspaper front page. The HEADLINE is huge and at the top — you see it in 1 second. The SUBHEADLINE gives supporting context. The ARTICLE BODY has details for those who want more. A dashboard follows the exact same hierarchy.

💡 Example

A sales dashboard layout following the 5-second rule:

- Top Centre (largest): Total Revenue Today — \$1.2M (green = above target)

- Top Left: Revenue vs. Target gauge chart
- Top Right: Top 3 performing products
- Middle: Regional sales heatmap
- Bottom: Full detailed data table with filters

Industry Use Case

Industry	5-Second Rule Application
Finance	Trading floor dashboards — P&L at the top so traders act in seconds
Healthcare	ICU monitoring — patient vitals most critical, placed top-centre
Operations	Supply chain dashboards — delayed shipments flagged at top in red
Marketing	Campaign dashboards — ROAS (return on ad spend) shown first, before breakdowns

1.3 Visual Perception: Gestalt Principles

Definition

Gestalt Principles are rules of visual perception that describe how the human brain naturally groups and interprets visual information. In dashboard design, applying these principles helps users understand data relationships instantly — without needing labels or explanations.

Gestalt is a German word meaning 'unified whole'. Our brain sees patterns before it reads text.

Sub-Topics

Principle 1 — Proximity

Things that are placed close together are perceived as belonging to the same group. Elements that are far apart are seen as separate.

- Dashboard application: Group related KPIs together (e.g., all Sales metrics in one area, all Operations metrics in another).
- Bad practice: Scattering related charts randomly across the dashboard — users don't know what belongs together.

Principle 2 — Similarity

Things that look alike (same color, shape, or size) are seen as related. Differences signal that things are different.

- Dashboard application: Use the same color (e.g., blue) for all revenue metrics, and a different color (e.g., orange) for cost metrics.
- Bad practice: Using random colors for every chart — no pattern, no meaning.

Principle 3 — Enclosure

Elements that are enclosed within a border, box, or background are perceived as a group, even if they are not touching.

- Dashboard application: Put a light shaded box or card around related charts to show they form one section.
- Bad practice: Placing unrelated charts next to each other with no visual separation — users think they are connected.

Principle 4 — Continuity

The eye follows smooth, continuous lines or curves naturally. This guides user attention along a path.

- Dashboard application: Use consistent horizontal rows of charts so the eye flows from left to right naturally.

Principle 5 — Figure/Ground

The brain separates objects from their background. The 'figure' (foreground) pops out; the 'ground' (background) recedes.

- Dashboard application: Use white backgrounds with coloured data points — data pops out clearly.

⚡ Why Important

- Gestalt principles reduce the need for legends, labels, and explanations — users 'just get it'.
- Misapplying these principles causes confusion — users misread relationships between charts.
- Well-designed dashboards apply all 3-5 principles simultaneously.

Real-World Analogy

Analogy Think of a restaurant menu. All appetizers are grouped together (Proximity), they all have the same font style (Similarity), and they are enclosed in a box section (Enclosure). You don't need a waiter to explain which items are appetizers — your brain knows instantly.

 Example

Applying all three core principles to a marketing dashboard:

- Proximity: Place all 'Social Media Metrics' (likes, shares, reach) in the top-left card together.
- Similarity: All positive metrics shown in green, all negative metrics in red — across every chart.
- Enclosure: A grey-bordered card groups 'Email Campaign Performance' charts, separate from social media charts.

 Industry Use Case

Industry	Gestalt Principle Application
Retail	Group basket size, conversion rate, avg. order value — all in a 'Sales Performance' card
SaaS	All churn-related metrics (churned users, churn rate, win-back rate) enclosed in a 'Retention' zone
Logistics	Delayed vs. on-time shipments shown in red vs. green (Similarity) across all charts
Finance	Risk metrics grouped separately from profitability metrics using visual enclosure

1.4 Color Theory: Using Color for Emphasis, Not Decoration

 Definition

Color Theory in dashboard design is the practice of using color deliberately to communicate meaning, guide attention, and distinguish data — NOT to make the dashboard look 'pretty'. Every color choice should have a purpose.

The golden rule: If you use color everywhere, it means nothing. Color only works when it is used sparingly.

 Sub-Topics

The 3 Roles of Color in Dashboards

Role	Purpose & Best Practice
Categorical Color	Distinguish between different groups (e.g., different product lines get different colors). Use max 5-7 distinct colors.
Sequential Color	Show intensity or magnitude — light to dark shading (e.g., a heatmap where deeper blue = higher sales).
Diverging Color	Show deviation from a centre point — green for above target, red for below target, white/grey for neutral/at target.

Traffic Light Convention

The most universally understood color system:

- Green = Good, on track, above target
- Yellow/Amber = Warning, approaching a threshold
- Red = Bad, below target, needs immediate attention

Color Accessibility — Colorblind-Safe Palettes

8% of men and 0.5% of women have color vision deficiency. Avoid red-green combinations as the primary way to distinguish data. Use:

- Blue vs. Orange as the main distinguishing pair (colorblind-safe)
- Patterns or shapes as a secondary encoding alongside color

Background and Contrast

- White or very light grey backgrounds work best for most dashboards.
- Dark themes (dark background, light text) work for operations/monitoring dashboards.
- Always ensure a contrast ratio of at least 4.5:1 for text legibility.

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⚡ Why Important

- Misused color causes 'rainbow chart syndrome' — too many colors, no meaning.
- Correct diverging colors instantly communicate performance vs. target at a glance.
- Ignoring accessibility alienates 8% of your audience.
- Consistent color usage across a report builds trust and reduces cognitive load.

🌐 Real-World Analogy

Analogy Think of a traffic light. It has only 3 colors, and each has a clear, universal meaning — you don't need instructions. A good dashboard uses color the same way: a few deliberate colors, each with a specific, consistent meaning. A bad dashboard is like a rainbow — beautiful, but meaningless.

 Example

A financial dashboard uses color correctly:

- Revenue metrics: Blue (neutral, informational)
- Above target performance: Dark green fill behind the number
- Below target performance: Red fill with white text
- Year-over-year comparison: Orange bars vs. blue bars
- Background: Clean white, with light grey card separators

Everything else — chart borders, axis lines, gridlines — uses very light grey so they don't compete for attention.

 Industry Use Case

Industry	Color Theory Application
Banking	Green/Red/Amber RAG (Red-Amber-Green) status for each loan portfolio segment
Supply Chain	Heatmap of delivery delays: light yellow = 1 day late, dark red = 5+ days late (sequential)
Healthcare	Patient risk scores: Blue (low risk), Yellow (moderate), Red (high risk) — traffic light system
SaaS	Blue = monthly active users, Orange = churned users — consistent across all charts in the report

1.5 Choosing the Right Chart: Comparison vs. Composition vs. Trend

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 Definition

Choosing the right chart means selecting the visualization type that most accurately and clearly represents what the data is actually saying. The wrong chart type creates confusion and misrepresents data, even if the numbers are correct.

The question to ask: 'What relationship am I trying to show?' Not 'What looks impressive?'

 Sub-Topics

The 4 Core Chart Categories

Category	When to Use It
Comparison	Comparing values between different groups or categories at a single point in time.
Trend / Time-Series	Showing how a metric changes over time.
Composition	Showing what parts make up a whole (part-to-whole relationships).
Relationship / Correlation	Showing whether two variables move together or are related.

Chart Type Selection Guide

Question You're Answering	Recommended Chart
Which region sold the most?	Bar Chart (horizontal for many categories)
How did revenue change month by month?	Line Chart
What % of revenue comes from each product?	Stacked Bar or Pie Chart (max 5 slices)
Is there a link between ad spend and sales?	Scatter Plot
How is a KPI performing vs. target?	Bullet Chart or Gauge
Comparing many metrics across categories?	Heatmap
Single most important number?	KPI Card / Big Number

Common Chart Mistakes to Avoid

- 3D Charts: Distort proportions — NEVER use 3D pie or 3D bar charts in professional dashboards.
- Pie Charts with 6+ slices: Human eyes cannot distinguish tiny slices — use a bar chart instead.
- Dual-Axis Charts: Two Y-axes on one chart confuse viewers — use two separate charts.
- Truncated Y-Axis: Starting the Y-axis at a non-zero value exaggerates differences.

 Why Important

- The wrong chart type can make small differences look massive (or vice versa) — misleading stakeholders.
- Choosing the right chart reduces the explanation time needed in meetings.
- It directly impacts whether decisions made from the dashboard are correct.

 Real-World Analogy

Analogy You wouldn't use a ruler to measure the temperature of water, even though both are measurement tools. Similarly, you wouldn't use a pie chart to show a trend over time, even though both are 'charts'. The tool must match the job. Using the wrong chart is as misleading as using the wrong measurement tool.

 Example

A marketing analyst needs to show three different things in one report:

- **COMPARISON:** Which of 5 campaigns got the most clicks? → Horizontal Bar Chart (easy to compare lengths)
- **TREND:** How did website traffic change over the last 12 months? → Line Chart (shows direction and momentum)
- **COMPOSITION:** What percentage of revenue came from each product category? → Stacked Bar Chart (shows part-to-whole AND comparison across months)

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 Industry Use Case

Industry	Chart Selection Example
Retail	Sales by store → Bar Chart Weekly sales trend → Line Chart Category revenue split → Stacked Bar
HR	Headcount by department → Bar Attrition trend 12 months → Line Tenure distribution → Histogram
Finance	Portfolio returns vs. benchmark → Line Asset allocation → Donut Chart Expense breakdown → Treemap

1.6 User Stories: Defining 'Who' the Dashboard Is For

Definition

A User Story is a short, simple description of what a specific type of user needs from the dashboard, written from their perspective. It forces the analyst to design for the audience — not for themselves.

Standard Format: As a [type of user], I want to [see/do something], so that I can [achieve a goal].

Sub-Topics

The Three Dashboard Audiences

Audience	Needs & Behavior
Executive / C-Suite	Wants high-level KPIs, no detail. Views dashboard for < 2 minutes. Needs RAG status at a glance.
Manager / Operations	Wants team or regional breakdowns. Needs to filter and compare. Spends 10-15 mins.
Analyst / Data Team	Wants full granularity, drill-down capability, raw numbers. Spends 30+ mins exploring.

Writing User Stories for Dashboards

Before building any chart, write user stories to define requirements:

- Executive story: As a CEO, I want to see total revenue vs. target this month, so that I can decide if we need an emergency sales push.
- Manager story: As a regional sales manager, I want to filter revenue by region and salesperson, so that I can identify who needs coaching.
- Analyst story: As a data analyst, I want to see daily transaction-level data with filters by product and channel, so that I can investigate anomalies.

The 5 W's of Dashboard Design

Question	What It Determines
Who is using it?	Level of detail, complexity, vocabulary used

What decisions do they make?	Which KPIs and charts to include
Where do they view it?	Screen size: desktop vs. mobile vs. TV wall
When do they view it?	Refresh frequency: real-time vs. daily vs. weekly
Why do they need it?	The core purpose — monitoring, analysis, or reporting

⚡ Why Important

- Without user stories, dashboards become feature dumps — everything is included, nothing is useful.
- User stories prevent 'build it and they won't come' — dashboards nobody looks at.
- They create alignment between analysts and stakeholders before a single chart is built.
- They give you a clear way to say 'no' to scope creep — if it doesn't serve a user story, it doesn't belong.

🌐 Real-World Analogy

Analogy A GPS app serves many users. The taxi driver needs real-time traffic and fastest route. The tourist needs points of interest and walking directions. The delivery driver needs multiple stops in sequence. One app, three completely different interfaces — all designed around specific user needs. Your dashboard is the same: know your user first, then design.

💡 Example

An HR department requests a 'People Analytics Dashboard'. The analyst writes user stories:

- As an HR Director, I want to see company-wide headcount and attrition rate, so that I can present workforce health to the board.
- As a Hiring Manager, I want to see open roles, time-to-fill, and candidate pipeline by department, so that I can plan my team's capacity.
- As an HR Analyst, I want to see raw employee data with filters by tenure, department, salary band, and performance rating, so that I can run ad hoc analysis.

Result: Three separate dashboard pages, each designed for one audience — not one crowded page trying to serve everyone.

🏭 Industry Use Case

Industry	User Story Application
Retail	CEO: total sales vs. last year Store Manager: store-level inventory & footfall Analyst: SKU-level sales data

SaaS	CEO: MRR & churn Product Manager: feature adoption rates Data Analyst: event-level clickstream
Healthcare	Hospital Director: bed occupancy % Ward Nurse: today's patient list Analyst: historical admission patterns
Finance	CFO: P&L summary Finance Manager: budget vs. actual by department Analyst: transaction-level ledger

1.7 Interactivity Design: Drill-downs, Tooltips & Page Navigation

Definition

Interactivity Design refers to planning and building features that allow users to actively engage with a dashboard — exploring data at different levels of depth without needing separate reports. The three main interactive features are Drill-downs, Tooltips, and Page Navigation.

Sub-Topics

Drill-Downs

A drill-down allows users to click on a high-level summary and expand into more granular detail automatically.

- Example: Click on 'North Region' in a map → bar chart updates to show individual city sales within the North Region.
- Drill-down hierarchy example: Country → State → City → Store → Product
- Best for: Exploring which sub-category is driving a top-level result.

Tooltips

A tooltip is a small information pop-up that appears when a user hovers over a data point, revealing additional context without cluttering the main chart.

- Example: Hover over a bar → shows exact value, % change vs. last year, and rank out of all bars.
- Design rule: Show 3-5 pieces of context in a tooltip maximum. More than that causes overload.
- Best for: Adding supplementary data that would make the main chart too busy if shown permanently.

Page Navigation / Bookmarks

Page navigation allows different user types to access different levels of the dashboard — from summary pages to detailed analytical pages.

- Common structure: Page 1 = Executive Summary | Page 2 = Operational Detail | Page 3 = Raw Data Explorer
- Bookmarks (in Power BI) or alternate views (in Tableau) allow saving specific filter states for quick access.
- Best for: Multi-audience dashboards where executives and analysts need different views of the same data.

Cross-Filtering / Slicers

When a user clicks or selects a filter value (e.g., 'Q1 2024'), ALL charts on the page update simultaneously to show data for that selection only.

- Slicers: Dropdown or button-based filters the user can control.
- Cross-filtering: Clicking a bar in one chart automatically filters all other charts on the page.

The Interactivity Decision Framework

User Need	Recommended Feature
I want to see details behind a summary number	Drill-down
I want extra context on a specific data point	Tooltip
I want to switch between summary and detail views	Page Navigation
I want to filter everything by one dimension	Slicer / Cross-filter
I want to save my current filtered view	Bookmark

⚡ Why Important

- Interactivity replaces the need for 20 separate static reports — one dashboard serves all exploration needs.
- It gives users independence — they don't need to ask analysts for every question.

- Poorly designed interactivity (too many drill-downs, confusing tooltips) actually frustrates users.

 Real-World Analogy

Analogy Think of Google Maps. You start with a world view (executive summary page). You zoom in to a country, then a city (drill-down). You hover over a restaurant and see its rating, hours, and distance (tooltip). You click 'Navigate' to switch to turn-by-turn mode (page navigation). Everything works together — you explore at your own pace.

 Example

A retail dashboard with full interactivity:

- Page 1 (Executive): Total revenue, top category, best region — big KPI cards. Click any card → goes to Page 2.
- Page 2 (Manager): Revenue bar chart by region. Click 'South' bar → drill-down shows South's cities.
- Tooltip on any bar: Exact revenue | % vs. target | Rank | % vs. last year.
- Slicer on right: Filter by Month, Product Category, or Channel — ALL charts update instantly.

 Industry Use Case

Industry	Interactivity Design Application
E-commerce	Drill from product category → product name → SKU. Tooltip shows inventory level + return rate.
Finance	Page 1: P&L summary. Page 2: Cost breakdown. Drill from department → cost centre → invoice level.
Telecom	Cross-filter: click a network tower on map → all charts update to show that tower's performance data
HR	Slicer for department + tenure band → headcount, attrition, and salary charts all update simultaneously

1.8 Handling 'Request Bloat': Saying No to Unnecessary Charts

 Definition

Request bloat is the phenomenon where stakeholders continuously add new charts, metrics, and features to a dashboard until it becomes unusable — a visual mess that serves no one well.

It happens because everyone believes their data is 'essential'. The analyst's job is to be a strategic editor, not just a developer.

A dashboard with 30 charts and 15 filters tells no story. A dashboard with 6 focused charts tells a clear story.

Sub-Topics

Why Request Bloat Happens

- Fear of missing out (FOMO): Stakeholders worry their data won't be seen if it's not on the dashboard.
- HiPPO effect: Highest Paid Person's Opinion — whoever has authority demands their metrics are included.
- No clear ownership: When many departments share a dashboard, everyone adds something.
- 'Just in case' thinking: 'We might need this chart someday' — even if it's never used.

The 3 Filter Tests for Every Requested Chart

Test	Question to Ask
Test 1: Decision Test	What specific decision will this chart help make? If no clear answer → reject.
Test 2: Audience Test	Which user story does this chart serve? If no user story covers it → reject.
Test 3: Frequency Test	How often will someone look at this chart? If 'rarely' → move to an appendix page, not the main dashboard.

Techniques for Saying No Professionally

- The 'Not Now' approach: 'That's a great idea — let's add it to Version 2 after we validate the core dashboard.'
- The Appendix approach: 'We'll put that on Page 3 (Data Explorer) so analysts can access it, but keep the main dashboard clean.'
- The Evidence approach: 'After the first month, we'll review which charts are actually being used and make decisions then.'
- The User Story redirect: 'Which of our defined user stories does this chart serve? Let's make sure it's in scope.'

What Belongs on a Dashboard vs. a Report

Dashboard	Report
Monitoring ongoing performance	One-time or periodic deep analysis
Max 6-12 KPIs in focus	Can have 30+ metrics
Updated automatically (live data)	Static snapshot in time
Scanned in seconds	Read in depth over minutes
Audience: Operational users	Audience: Analytical users

⚡ Why Important

- Bloated dashboards have lower adoption rates — users stop looking at them because they're overwhelming.
- Every chart added is cognitive load added — there is a real cost to adding 'just one more'.
- A clean, focused dashboard with 8 charts that people trust is worth more than 40 charts that confuse everyone.
- Saying no professionally is a core analytical skill that protects your credibility and the dashboard's purpose.

🌐 Real-World Analogy

Analogy Think of a cockpit of a commercial airplane. Pilots need to monitor hundreds of data points — but the dashboard shows only the 10-15 most critical ones prominently. Everything else is available but hidden unless needed. Engineers didn't add every possible sensor to the main display — they made hard choices about what the pilot needs to see in real-time. Your dashboard needs the same discipline.

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💡 Example

A stakeholder requests 12 new charts for a sales dashboard. The analyst applies the three tests:

- ☒ 'Revenue by product category' → passes Decision Test + is in Manager user story → INCLUDE
- ☒ 'Sales pipeline conversion rate' → passes all 3 tests → INCLUDE
- ☐ 'Number of emails sent by sales reps' → fails Decision Test (no actionable decision tied to it) → MOVE TO APPENDIX

- ✗ 'Revenue in currencies of all 23 countries we operate in' → used by 1 analyst, never by managers or execs → REJECT from main dashboard, link to separate report

Industry Use Case

Industry	Request Bloat Handling Example
E-commerce	CEO requests daily chart of returns by colour variant — fails Audience Test (operational, not strategic). Redirected to ops report.
Banking	Compliance team wants 15 risk metrics on the CEO dashboard — analyst negotiates a single 'Risk Health Score' KPI card instead.
Healthcare	Each department wants their own KPIs on the main board — analyst creates dept.-specific sub-pages, keeps main dashboard to 8 hospital-wide KPIs.
Marketing	Agency requests chart for every ad creative's performance — redirected to a dedicated creative analytics page, not the main campaign dashboard.

Quick Reference Summary

Topic	Core Takeaway
1.1 KPIs	Use BOTH lead (predictive) and lag (resultant) indicators on every dashboard.
1.2 Five-Second Rule	Most critical info must be understood in 5 seconds — use layout hierarchy.
1.3 Gestalt Principles	Group related elements (Proximity), use consistent styles (Similarity), section with borders (Enclosure).
1.4 Color Theory	Color = meaning, not decoration. Traffic lights, diverging scales, max 5-7 colors.
1.5 Chart Selection	Match chart type to the relationship: Comparison, Trend, Composition, or Correlation.
1.6 User Stories	Design for a specific person, not for 'everyone'. Know the Who before the What.
1.7 Interactivity	Use drill-downs for depth, tooltips for context, page navigation for audiences.
1.8 Request Bloat	Every chart must pass Decision + Audience + Frequency tests. Say no professionally.

INTERVIEW-BASE UNIT TEST

Unit 1: Dashboard Design & UX Strategy

Business Intelligence & Dashboards

⚡ LEVEL 1: WARMUP TEST — 5 MCQ Questions

Instructions: Choose the single best answer for each question.

Q1. Which of the following is a LEAD indicator?

- A) Total revenue earned last quarter
- B) Number of sales calls made this week
- C) Annual profit margin
- D) Customer churn rate last month

Q2. What does the "5-Second Rule" in dashboard design state?

- A) A dashboard must load in 5 seconds
- B) Users should click a chart within 5 seconds
- C) The most important insight should be understood within 5 seconds of viewing
- D) Tooltips should appear within 5 seconds of hovering

Q3. Which Gestalt Principle states that elements enclosed within a border are perceived as a group?

- A) Proximity
- B) Similarity
- C) Continuity
- D) Enclosure

Q4. Which chart type is MOST appropriate to show how revenue changed month by month over a year?

- A) Pie Chart
- B) Scatter Plot
- C) Line Chart
- D) Heatmap

Q5. What is the standard format for writing a User Story in dashboard design?

- A) "The dashboard should show [metric] for [user]."
- B) "As a [type of user], I want to [action], so that I can [goal]."
- C) "When [condition], display [chart] to [user]."
- D) "Build [feature] for [user] because [reason]."

LEVEL 2: INTERVIEWER TEST — 15 Intermediate Semi-Advance Questions

Answer all questions in a conversational, thoughtful student format — acknowledge the "why" behind concepts, keep tone practical, and include brief code examples where relevant.

Section A: Descriptive Questions (4 Questions)

Q1. Explain the difference between Lead and Lag Indicators. Why is it important to include BOTH types on a business intelligence dashboard? Provide examples from two different industries.

 **Answer**

Q2. Describe all five Gestalt Principles relevant to dashboard design. For each principle, explain how misapplying it can negatively affect user experience.

 **Answer**

Q3. What is Color Theory in the context of dashboard design? Explain the three roles of color (Categorical, Sequential, Diverging) with real-world dashboard examples for each.

☒ Answer

Q4. Explain the concept of "Request Bloat" in dashboard design. What causes it, and what professional techniques can an analyst use to say no while maintaining stakeholder relationships?

☒ Answer

Section B: Scenario-Based Descriptive Questions (4 Questions)

Q1. SCENARIO: A hospital's operations team asks you to build a single dashboard for three users: the Hospital Director (needs bed occupancy % at a glance), Ward Nurses (need today's patient list by ward), and a Health Data Analyst (needs historical admission patterns with filters). How would you approach this, and what interactivity features would you use?

☒ Answer

Q2. SCENARIO: You have designed a clean 8-chart sales dashboard. A senior VP storms into your meeting and demands you add 7 more charts immediately — including a map of revenue by country for all 50 countries, hourly revenue fluctuations, and a chart showing revenue in 23 different currencies. How do you respond?

☒ **Answer**

Q3. SCENARIO: An e-commerce analyst is building a campaign performance dashboard. The marketing team wants to compare 5 campaigns' total clicks, show how daily website traffic changed over 30 days, AND show what % of revenue came from each of 4 product categories. Which chart types would you use for each, and why?

☒ **Answer**

Q4. SCENARIO: You are reviewing a colleague's dashboard and notice: a 3D pie chart with 9 slices, a dual-axis chart mixing revenue and profit %, and a bar chart with a Y-axis starting at 850,000 instead of 0. Identify all design violations, explain why each is problematic, and propose the correct solution.

✓ Answer

Section C: Coding Questions (4 Questions)

Q1. Using Python and pandas, write code to calculate both a Lead KPI (7-day rolling average of daily active users) and a Lag KPI (monthly total revenue). The dataset has columns: date, daily_active_users, and revenue.

✓ Answer

Understanding the "why" here matters: rolling averages smooth out daily noise so you can see the trend signal clearly — that's what makes it useful as a lead indicator. Monthly total is a clean lag confirmation.

Code:

✓ Answer

Q2. Write a Python function that evaluates a KPI value against a target and returns a color-coded RAG (Red/Amber/Green) status. The function should accept: value, target, and an amber_threshold (default 10%). Apply this to a list of 5 KPIs and print a styled summary.

✓ Answer

The RAG pattern is foundational to dashboard color theory — it implements the diverging color role (green = above, amber = warning, red = below). The threshold logic is where the real value is.

Code:

✓ Answer

Q3. Using Python and matplotlib, write code to create a properly designed dashboard chart panel that demonstrates chart selection best practices: a bar chart for comparison, a line chart for trends, and a donut chart for composition — all in one figure with correct formatting.

✓ Answer

Chart selection best practice says: match the chart to the relationship being shown. This code demonstrates all three categories in one figure.

Code:

✓ Answer

Q4. Write a Python function called `apply_5_second_rule()` that takes a list of KPI dictionaries (with name, value, target, and priority fields) and sorts them by priority, flags any that fail a simple threshold test, and returns a formatted layout plan showing Zone 1 (top-left), Zone 2 (top-right), and Zone 3 (bottom) assignments.

✓ Answer

This is the 5-Second Rule and Information Hierarchy implemented in code — the most critical KPI gets Zone 1 automatically based on priority and performance status.

Code:

✓ Answer

Section D: Database Scenario-Based Questions (3 Questions)

Q1. DATABASE SCENARIO: You have a `sales_transactions` table with columns: `transaction_id`, `salesperson_id`, `region`, `sale_date`, `revenue`, and `target_revenue`. Write SQL to generate a KPI summary report showing: total actual revenue, total target revenue, attainment %, and RAG status (Green $\geq 100\%$, Amber $\geq 90\%$, Red $< 90\%$) — grouped by region. Sort by attainment ascending (worst-performing regions first).

✓ Answer

This is a classic dashboard backend query — grouping real vs. target data and applying business logic (RAG classification) inside SQL. Sorting worst-first is intentional: it ensures problem regions are visible immediately on the dashboard.

SQL:

✓ Answer

Q2. DATABASE SCENARIO: You are building an interactivity feature — a drill-down from monthly revenue to weekly breakdown. The `dashboard_sales` table has: `sale_id`, `sale_date`, `region`, `product_category`, `revenue`. Write two SQL queries: Query 1 returns the monthly revenue summary. Query 2 accepts a month parameter and returns weekly revenue for that month (as a drill-down result).

✓ Answer

This is the SQL backbone of a drill-down interaction — the first query populates the summary chart, and clicking a bar triggers the second query with the selected month as a parameter. Understanding this connection is key to building interactive dashboards.

Query 1 — Monthly Revenue Summary (Summary Layer):

✓ Answer

Query 2 — Weekly Drill-Down (accepts month parameter e.g. '2024-03'):

✓ Answer

Q3. DATABASE SCENARIO: To prevent request bloat, you are asked to build a "dashboard usage analytics" table to track which charts are actually being viewed. Design the schema for a `chart_view_log` table and write a SQL query that identifies "zombie charts" — charts that have had zero views in the last 30 days — so they can be removed from the dashboard.

✓ Answer

This is a meta-analytics problem — using data to make decisions ABOUT the dashboard itself. Finding zombie charts is the evidence-based approach to fighting request bloat without needing subjective debates with stakeholders.

Schema Design:

✓ Answer

Zombie Chart Detection Query:

☒ Answer



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INTERVIEW-BASE UNIT TEST

Unit 1: Dashboard Design & UX Strategy

Business Intelligence & Dashboards

⚡ LEVEL 1: WARMUP TEST — 5 MCQ Questions

Instructions: Choose the single best answer for each question.

Q1. Which of the following is a LEAD indicator?

- A) Total revenue earned last quarter
- B) Number of sales calls made this week ✓
- C) Annual profit margin
- D) Customer churn rate last month

Q2. What does the "5-Second Rule" in dashboard design state?

- A) A dashboard must load in 5 seconds
- B) Users should click a chart within 5 seconds
- C) The most important insight should be understood within 5 seconds of viewing ✓
- D) Tooltips should appear within 5 seconds of hovering

Q3. Which Gestalt Principle states that elements enclosed within a border are perceived as a group?

- A) Proximity
- B) Similarity
- C) Continuity
- D) Enclosure ✓

Q4. Which chart type is MOST appropriate to show how revenue changed month by month over a year?

- A) Pie Chart
- B) Scatter Plot
- C) Line Chart ✓
- D) Heatmap

Q5. What is the standard format for writing a User Story in dashboard design?

- A) "The dashboard should show [metric] for [user]."
- B) "As a [type of user], I want to [action], so that I can [goal]." ✓
- C) "When [condition], display [chart] to [user]."
- D) "Build [feature] for [user] because [reason]."

🎯 LEVEL 2: INTERVIEWER TEST — 15 Intermediate Semi-Advance Questions

Answer all questions in a conversational, thoughtful student format — acknowledge the "why" behind concepts, keep tone practical, and include brief code examples where relevant.

📄 Section A: Descriptive Questions (4 Questions)

Q1. Explain the difference between Lead and Lag Indicators. Why is it important to include BOTH types on a business intelligence dashboard? Provide examples from two different industries.

✅ Answer

Great question — this is something I find really intuitive once you frame it correctly.

Lead indicators are forward-looking metrics you can act on RIGHT NOW. They give early warning signals before an outcome is confirmed. For example, the number of sales calls made today is a lead indicator — you can increase calls to impact future revenue.

Lag indicators are backward-looking results that confirm what already happened. Total revenue earned last quarter is a lag — it tells you the outcome, but you cannot change it.

The reason BOTH matter on a dashboard is this: if you only show lag indicators, you are always reacting — like reading yesterday's newspaper to make today's decisions. If you only show lead indicators, you don't know whether your efforts actually worked.

Industry Example 1 (Healthcare): Lead = Patient wait time today (you can assign more staff now).
Lag = Monthly patient satisfaction score (confirms whether efforts worked).

Industry Example 2 (Marketing): Lead = Email open rate this campaign (signal of engagement). Lag = Customer acquisition cost last quarter (confirms campaign ROI).

A smart dashboard always pairs a predictive metric with its confirmation metric — so decision-makers can both anticipate and verify performance.

Q2. Describe all five Gestalt Principles relevant to dashboard design. For each principle, explain how misapplying it can negatively affect user experience.

✓ Answer

Gestalt principles are fascinating because they describe how our brains are wired — we perceive patterns before we read text. Understanding these is fundamental to design that feels intuitive.

1. Proximity: Elements placed close together are seen as a group. Misapplication: Scattering related charts randomly confuses users about which metrics belong together — they spend mental energy figuring out structure instead of reading data.

2. Similarity: Things that look alike (same color/shape) are seen as related. Misapplication: Using random colors for each chart destroys meaning — users cannot identify which metrics share a category.

3. Enclosure: Elements within a border/box are perceived as a group. Misapplication: Placing unrelated charts side-by-side without visual separation creates false relationships — users assume they are linked.

4. Continuity: The eye follows smooth, continuous lines naturally. Misapplication: Inconsistent row/column alignment forces the eye to jump unnaturally, increasing cognitive fatigue.

5. Figure/Ground: The brain separates objects from their background. Misapplication: A dark or busy background competes with data points — the "figure" (data) doesn't pop, reducing readability.

Well-designed dashboards apply multiple principles simultaneously — this is why great design feels effortless while poor design feels exhausting.

Q3. What is Color Theory in the context of dashboard design? Explain the three roles of color (Categorical, Sequential, Diverging) with real-world dashboard examples for each.

✓ Answer

Color theory in dashboards is really about discipline — using color with purpose rather than decoration. The golden rule I always keep in mind: if everything is highlighted, nothing is highlighted.

1. Categorical Color: Used to distinguish between different groups. Example: A retail dashboard uses blue for Product A revenue bars and orange for Product B revenue bars — different colors, different categories. Maximum 5-7 distinct colors; beyond that, viewers cannot keep track.

2. Sequential Color: Used to show intensity or magnitude through light-to-dark shading. Example: A supply chain heatmap where light yellow = 1 day delayed, medium orange = 3 days, dark red = 5+ days delayed. The shade itself communicates severity without a legend.

3. Diverging Color: Used to show deviation from a center point. Example: A financial dashboard showing sales vs. target — dark green for above target, white/grey for at target, dark red for below target. This immediately communicates performance direction.

Additional consideration: 8% of men have color vision deficiency, so blue-orange is a safer distinguishing pair than red-green. Accessibility is not optional in professional dashboard design.

Q4. Explain the concept of "Request Bloat" in dashboard design. What causes it, and what professional techniques can an analyst use to say no while maintaining stakeholder relationships?

 Answer

Request bloat is something every analyst eventually faces — it's when stakeholders keep adding charts, metrics, and features until the dashboard collapses under its own complexity. A dashboard with 30 charts tells no story.

Why it happens: The HiPPO effect (Highest Paid Person's Opinion), FOMO from departments worried their data won't be visible, no clear ownership of the dashboard, and "just in case" thinking where metrics are added speculatively.

The 3-test filter I use for every requested chart:

- Decision Test: What decision does this chart enable? No clear answer → reject.
- Audience Test: Which user story does this serve? Not in scope → reject.
- Frequency Test: How often will someone look at this? Rarely → move to an appendix page.

Professional techniques for saying no:

- "Not Now" approach: "That's a great idea for Version 2 — let's validate the core first."
- Appendix approach: "We'll put this on Page 3 for analysts — keeps the main dashboard clean."
- Evidence approach: "After the first month, we'll review usage analytics and revisit this."

The key insight is that saying no professionally actually builds credibility — stakeholders learn to trust that what IS on the dashboard truly matters.

Section B: Scenario-Based Descriptive Questions (4 Questions)

Q1. SCENARIO: A hospital's operations team asks you to build a single dashboard for three users: the Hospital Director (needs bed occupancy % at a glance), Ward Nurses (need today's patient list by ward), and a Health Data Analyst (needs historical admission patterns with filters). How would you approach this, and what interactivity features would you use?

 Answer

This is a classic multi-audience problem, and the answer is that one page cannot serve all three users effectively. Here's how I'd approach it:

User Story Strategy: I'd write three formal user stories first — one for each persona — to define exactly what each user needs before designing anything.

Page structure using interactivity:

- Page 1 — Executive View: Hospital Director. KPI card showing Bed Occupancy % (large, top-center). RAG (Red/Amber/Green) color coding. Updated daily. No filters — just the key number with a trend sparkline.
- Page 2 — Operational View: Ward Nurses. Table showing today's patient list. Slicer to filter by Ward. Cross-filtering between ward map and patient list. Refreshed every 4 hours or in near-real-time.
- Page 3 — Analytical View: Data Analyst. Full historical admission data with date range slicers, department filters, diagnosis filters. Drill-down from ward level → individual patient admission records. Export capability.

Navigation Design: Clear page tabs or a home page with three role-based buttons: "Hospital Director View", "Ward Operations", "Analyst Explorer". Each button uses bookmarks/page navigation to route the right person to the right page.

The interactivity features I'd prioritize: drill-downs (ward → patient), slicers (department, date), cross-filtering, and tooltips showing bed status, days occupied, and ward capacity. This turns one dataset into three purpose-built experiences.

Q2. SCENARIO: You have designed a clean 8-chart sales dashboard. A senior VP storms into your meeting and demands you add 7 more charts immediately — including a map of revenue by country for all 50 countries, hourly revenue fluctuations, and a chart showing revenue in 23 different currencies. How do you respond?

✓ Answer

This is the HiPPO (Highest Paid Person's Opinion) problem in action — and it's a situation that tests both technical judgment and communication skills. My response would be structured, not defensive.

First, I'd acknowledge the intent: "These are genuinely interesting data points — let me think through where each of these fits best for your needs."

Then I'd apply the 3-filter test openly:

- Revenue map across 50 countries: "A 50-country map at this scale is visually unreadable — dots overlap and the chart loses meaning. Instead, I'd suggest a top 10 countries bar chart, with a 'View All' drill-down page for the full breakdown. This gives you the headline insight instantly."
- Hourly revenue fluctuations: "This serves our real-time operations team — I'd recommend this on a separate operations sub-page or a dedicated live monitoring dashboard. It would overwhelm an executive sales view."
- Revenue in 23 currencies: "This is an analyst-level requirement used by 1-2 finance team members. I'd build this as a linked report accessible from the dashboard, not on the main dashboard page itself."

The framing I'd use with the VP: "My goal is to make sure you can read the dashboard in 5 seconds and trust every number on it. Adding 7 more charts will make that impossible — but I can build a connected Page 2 that houses all this detail for your team."

This protects the dashboard's integrity while showing the VP their needs will be met — just in the right place.

Q3. SCENARIO: An e-commerce analyst is building a campaign performance dashboard. The marketing team wants to compare 5 campaigns' total clicks, show how daily website traffic changed over 30 days, AND show what % of revenue came from each of 4 product categories. Which chart types would you use for each, and why?

 Answer

This is a classic chart selection problem — three different "questions" requiring three different chart types.

Requirement 1 — Compare 5 campaigns' total clicks: Use a HORIZONTAL BAR CHART. Why? This is a comparison question (which is bigger/smaller?). Horizontal bars are ideal when you have 5+ categories because labels fit naturally on the left axis without rotation. The eye compares bar lengths intuitively — much cleaner than a vertical bar for many categories.

Requirement 2 — How daily traffic changed over 30 days: Use a LINE CHART. Why? This is a trend/time-series question. Line charts are the right tool when you want to show direction, momentum, and change over time. The connecting line between daily points helps the eye follow the trajectory — a bar chart for 30 days looks cluttered and doesn't communicate trend as clearly.

Requirement 3 — Revenue % from 4 product categories: Use a PIE CHART or DONUT CHART (4 slices is within the maximum of 5). Why? This is a composition question (part-to-whole). With only 4 categories, slices are large enough to be distinguishable. I'd choose a donut chart specifically — it provides a clean center space for the total revenue figure, adding extra context.

If the team also wanted to compare the category revenue split across months, I'd upgrade from pie to a STACKED BAR CHART — it handles both composition AND comparison across time periods simultaneously.

Q4. SCENARIO: You are reviewing a colleague's dashboard and notice: a 3D pie chart with 9 slices, a dual-axis chart mixing revenue and profit %, and a bar chart with a Y-axis starting at 850,000 instead of 0. Identify all design violations, explain why each is problematic, and propose the correct solution.

 Answer

This is a textbook example of three separate anti-patterns — I'd address each one clearly in a design review.

Violation 1 — 3D Pie Chart with 9 slices: Problem: 3D perspective distorts proportions — slices at the "front" of the chart appear larger than those at the "back", even if they have the same value. This actively misleads viewers. 9 slices is also far beyond the maximum of 5 — human eyes cannot

distinguish tiny slices reliably. Solution: Replace with a horizontal bar chart sorted by value (largest to smallest). This handles unlimited categories and allows precise length comparison.

Violation 2 — Dual-axis chart (Revenue + Profit %): Problem: Two Y-axes on one chart create cognitive overload. Viewers must constantly look back and forth to understand which axis corresponds to which line, and the visual relationship between the two lines is misleading — small changes on one axis look equivalent to large changes on the other. Solution: Split into two separate single-axis charts placed side by side. Viewers can compare them contextually without confusion.

Violation 3 — Y-axis starting at 850,000 (truncated): Problem: A truncated Y-axis dramatically exaggerates differences. What might be a 2% variance in revenue appears as a massive cliff on screen, manufacturing urgency that doesn't exist in the data. This is a data integrity issue, not just an aesthetic one. Solution: Set the Y-axis to start at 0, unless there is a documented, explicit reason to truncate (and even then, clearly label the truncation with a break symbol on the axis).

I'd communicate this to my colleague as: "Three quick fixes that will make this dashboard more trustworthy — let me show you the corrected versions and the reasoning behind each."

Section C: Coding Questions (4 Questions)

Q1. Using Python and pandas, write code to calculate both a Lead KPI (7-day rolling average of daily active users) and a Lag KPI (monthly total revenue). The dataset has columns: date, daily_active_users, and revenue.

Answer

Understanding the "why" here matters: rolling averages smooth out daily noise so you can see the trend signal clearly — that's what makes it useful as a lead indicator. Monthly total is a clean lag confirmation.

Code:

```
import pandas as pd

# Sample dataset
data = {
    "date": pd.date_range(start="2024-01-01", periods=90, freq="D"),
    "daily_active_users": [200 + i*2 for i in range(90)],
    "revenue": [5000 + i*100 for i in range(90)]
}
df = pd.DataFrame(data)
df["date"] = pd.to_datetime(df["date"])

# — LEAD INDICATOR: 7-Day Rolling Average of Daily Active Users —
df["rolling_avg_dau"] = df["daily_active_users"].rolling(window=7).mean()
```

```
# — LAG INDICATOR: Monthly Total Revenue —
df["month"] = df["date"].dt.to_period("M")
monthly_revenue = df.groupby("month")["revenue"].sum().reset_index()
monthly_revenue.columns = ["month", "total_monthly_revenue"]

# — Output —
print("Lead Indicator — 7-Day Rolling Avg DAU:")
print(df[["date", "daily_active_users", "rolling_avg_dau"]].tail(10))

print("\nLag Indicator — Monthly Total Revenue:")
print(monthly_revenue)
```

Q2. Write a Python function that evaluates a KPI value against a target and returns a color-coded RAG (Red/Amber/Green) status. The function should accept: value, target, and an amber_threshold (default 10%). Apply this to a list of 5 KPIs and print a styled summary.

✓ Answer

The RAG pattern is foundational to dashboard color theory — it implements the diverging color role (green = above, amber = warning, red = below). The threshold logic is where the real value is.

Code:

```
def rag_status(value, target, amber_threshold=0.10):
    """
    Returns RAG status for a KPI.
    value: actual KPI value
    target: expected target value
    amber_threshold: % below target before Amber fires (default 10%)
    """
    if target == 0:
        return "GREY", "No target defined"

    performance_ratio = value / target

    if performance_ratio >= 1.0:
        return "GREEN", f"{value} ≥ target {target} ({performance_ratio*100:.1f}%)"
    elif performance_ratio >= (1 - amber_threshold):
        return "AMBER", f"{value} within {amber_threshold*100:.0f}% of target {target}"
    else:
        return "RED", f"{value} significantly below target {target}"

# — Apply to 5 KPIs —
kpis = [
    {"name": "Monthly Revenue", "value": 120000, "target": 100000},
    {"name": "Customer Satisfaction", "value": 88, "target": 90},
    {"name": "Sales Conversion Rate", "value": 4.2, "target": 5.0},
```

```

{"name": "NPS Score", "value": 72, "target": 70},
{"name": "Churn Rate (lower=good)", "value": 6.5, "target": 5.0},
]

print(f"{'KPI':<30} {'VALUE':>10} {'TARGET':>10} {'STATUS':<8} DETAIL")
print("-" * 80)
for kpi in kpis:
    status, detail = rag_status(kpi["value"], kpi["target"])
    icon = "☑" if status == "GREEN" else "⚠" if status == "AMBER" else "✖"
    print(f"{kpi['name']:<30} {str(kpi['value']):>10} {str(kpi['target']):>10} {icon} {status:<6} {detail}")

```

Q3. Using Python and matplotlib, write code to create a properly designed dashboard chart panel that demonstrates chart selection best practices: a bar chart for comparison, a line chart for trends, and a donut chart for composition — all in one figure with correct formatting.

☑ Answer

Chart selection best practice says: match the chart to the relationship being shown. This code demonstrates all three categories in one figure.

Code:

```

import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import numpy as np

fig, axes = plt.subplots(1, 3, figsize=(16, 5))
fig.suptitle("Dashboard Chart Selection Best Practices", fontsize=14,
fontweight="bold")

# — Chart 1: BAR CHART — Comparison —
campaigns = ["Email", "Paid Search", "Social", "Referral", "Direct"]
clicks = [12000, 9500, 7800, 4300, 3100]
colors = ["#2E75B6" if c == max(clicks) else "#8CB4DC" for c in clicks]
axes[0].barh(campaigns, clicks, color=colors)
axes[0].set_title("Campaign Clicks (Comparison)", fontweight="bold")
axes[0].set_xlabel("Total Clicks")
axes[0].invert_yaxis() # Largest bar at top
axes[0].spines[["top", "right"]].set_visible(False)

# — Chart 2: LINE CHART — Trend/Time-Series —
days = list(range(1, 31))
traffic = [1200 + np.random.randint(-100, 200) + d*20 for d in days]
axes[1].plot(days, traffic, color="#2E75B6", linewidth=2, marker="o",
markersize=3)
axes[1].fill_between(days, traffic, alpha=0.1, color="#2E75B6")
axes[1].set_title("Daily Website Traffic — 30 Days (Trend)",
fontweight="bold")
axes[1].set_xlabel("Day")

```

```

axes[1].set_ylabel("Visitors")
axes[1].spines[["top", "right"]].set_visible(False)

# — Chart 3: DONUT CHART – Composition —
categories = ["Electronics", "Clothing", "Home", "Sports"]
revenue_pct = [38, 27, 21, 14]
palette = ["#2E75B6", "#F4A261", "#2A9D8F", "#E76F51"]
wedges, texts, autotexts = axes[2].pie(
    revenue_pct, labels=categories, colors=palette,
    autopct="%1.0f%%", startangle=90,
    wedgeprops=dict(width=0.55) # Donut hole
)
axes[2].set_title("Revenue by Category (Composition)", fontweight="bold")

plt.tight_layout()
plt.savefig("dashboard_charts.png", dpi=150, bbox_inches="tight")
plt.show()
print("Chart saved as dashboard_charts.png")

```

Q4. Write a Python function called `apply_5_second_rule()` that takes a list of KPI dictionaries (with name, value, target, and priority fields) and sorts them by priority, flags any that fail a simple threshold test, and returns a formatted layout plan showing Zone 1 (top-left), Zone 2 (top-right), and Zone 3 (bottom) assignments.

✓ Answer

This is the 5-Second Rule and Information Hierarchy implemented in code — the most critical KPI gets Zone 1 automatically based on priority and performance status.

Code:

```

def apply_5_second_rule(kpis):
    """
    Takes a list of KPI dicts and assigns them to dashboard zones
    based on the 5-Second Rule (most critical info gets Zone 1).
    Each KPI dict: {name, value, target, priority (1=highest)}
    """
    def rag(v, t):
        ratio = v / t if t else 0
        if ratio >= 1.0: return "GREEN"
        elif ratio >= 0.90: return "AMBER"
        return "RED"

    # Annotate each KPI with RAG and sort by priority
    for k in kpis:
        k["rag"] = rag(k["value"], k["target"])
        k["pct"] = round((k["value"] / k["target"]) * 100, 1) if k["target"] else 0

    # Sort by priority (1 = most important), then RED first within same priority
    rag_order = {"RED": 0, "AMBER": 1, "GREEN": 2}

```

```

sorted_kpis = sorted(kpis, key=lambda k: (k["priority"],
rag_order[k["rag"]]))

# Zone assignment
zones = {"Zone 1 (Top-Left - PRIMARY)": [],
"Zone 2 (Top-Right - SECONDARY)": [],
"Zone 3 (Bottom - TERTIARY)": []}

for i, kpi in enumerate(sorted_kpis):
    if i == 0:
        zones["Zone 1 (Top-Left - PRIMARY)"].append(kpi)
    elif i <= 2:
        zones["Zone 2 (Top-Right - SECONDARY)"].append(kpi)
    else:
        zones["Zone 3 (Bottom - TERTIARY)"].append(kpi)

# Print layout plan
print("=" * 60)
print("DASHBOARD LAYOUT PLAN - 5-Second Rule Applied")
print("=" * 60)
for zone, items in zones.items():
    print(f"\n{zone}:")
    for k in items:
        flag = "🟢" if k["rag"]=="RED" else "🟡" if k["rag"]=="AMBER" else "🟠"
        print(f" {flag} {k["name"]}: {k["value"]} / {k["target"]} ({k["pct"]} % of
target)")
    return zones

# — Test —
kpis = [
{"name": "Total Revenue", "value": 95000, "target": 100000, "priority": 1},
{"name": "Customer NPS", "value": 72, "target": 75, "priority": 2},
{"name": "Churn Rate", "value": 6.0, "target": 5.0, "priority": 2},
{"name": "New Signups", "value": 430, "target": 400, "priority": 3},
{"name": "Support Tickets", "value": 210, "target": 150, "priority": 3},
]
apply_5_second_rule(kpis)

```

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Section D: Database Scenario-Based Questions (3 Questions)

Q1. DATABASE SCENARIO: You have a sales_transactions table with columns: transaction_id, salesperson_id, region, sale_date, revenue, and target_revenue. Write SQL to generate a KPI summary report showing: total actual revenue, total target revenue, attainment %, and RAG status (Green ≥ 100%, Amber ≥ 90%, Red < 90%) — grouped by region. Sort by attainment ascending (worst-performing regions first).

✓ Answer

This is a classic dashboard backend query — grouping real vs. target data and applying business logic (RAG classification) inside SQL. Sorting worst-first is intentional: it ensures problem regions are visible immediately on the dashboard.

SQL:

```
SELECT
  region,
  SUM(revenue) AS total_actual_revenue,
  SUM(target_revenue) AS total_target_revenue,
  ROUND(
    SUM(revenue) * 100.0 / NULLIF(SUM(target_revenue), 0),
    2) AS attainment_pct,
  CASE
    WHEN SUM(revenue) >= SUM(target_revenue)
    THEN 'GREEN — On/Above Target'
    WHEN SUM(revenue) >= SUM(target_revenue) * 0.90
    THEN 'AMBER — Within 10% of Target'
    ELSE
    'RED — Below 90% of Target'
  END AS rag_status
FROM
  sales_transactions
GROUP BY
  region
ORDER BY
  attainment_pct ASC; -- Worst performers first (critical for dashboard design)

-- NULLIF prevents division-by-zero when target is 0
-- Multiply by 100.0 (not 100) to force decimal division in integer DBs
```

Q2. DATABASE SCENARIO: You are building an interactivity feature — a drill-down from monthly revenue to weekly breakdown. The dashboard_sales table has: sale_id, sale_date, region, product_category, revenue. Write two SQL queries: Query 1 returns the monthly revenue summary. Query 2 accepts a month parameter and returns weekly revenue for that month (as a drill-down result).

✓ Answer

This is the SQL backbone of a drill-down interaction — the first query populates the summary chart, and clicking a bar triggers the second query with the selected month as a parameter. Understanding this connection is key to building interactive dashboards.

Query 1 — Monthly Revenue Summary (Summary Layer):

```
-- Query 1: Monthly Revenue Summary
SELECT
  DATE_FORMAT(sale_date, '%Y-%m') AS month,
  COUNT(sale_id) AS total_transactions,
```



```

ROUND(SUM(revenue), 2) AS total_revenue
FROM
  dashboard_sales
GROUP BY
  DATE_FORMAT(sale_date, '%Y-%m')
ORDER BY
  month ASC;

```

Query 2 — Weekly Drill-Down (accepts month parameter e.g. '2024-03'):

```

-- Query 2: Weekly Revenue Drill-Down for selected month
-- Replace '2024-03' with the dashboard parameter value
SELECT
  WEEK(sale_date, 1) AS week_number,
  MIN(sale_date) AS week_start,
  MAX(sale_date) AS week_end,
  COUNT(sale_id) AS transactions,
  ROUND(SUM(revenue), 2) AS weekly_revenue
FROM
  dashboard_sales
WHERE
  DATE_FORMAT(sale_date, '%Y-%m') = '2024-03' -- Passed from dashboard click
GROUP BY
  WEEK(sale_date, 1)
ORDER BY
  week_number ASC;

-- Drill-down hierarchy: Monthly → Weekly → Daily → Transaction-level

```

Q3. DATABASE SCENARIO: To prevent request bloat, you are asked to build a "dashboard usage analytics" table to track which charts are actually being viewed. Design the schema for a chart_view_log table and write a SQL query that identifies "zombie charts" — charts that have had zero views in the last 30 days — so they can be removed from the dashboard.

Answer

This is a meta-analytics problem — using data to make decisions ABOUT the dashboard itself. Finding zombie charts is the evidence-based approach to fighting request bloat without needing subjective debates with stakeholders.

Schema Design:

```

-- Table 1: Registry of all charts on all dashboards
CREATE TABLE dashboard_charts (
  chart_id INT PRIMARY KEY AUTO_INCREMENT,
  dashboard_name VARCHAR(100) NOT NULL,
  chart_name VARCHAR(150) NOT NULL,
  chart_type VARCHAR(50), -- bar, line, pie, kpi_card, etc.
  added_date DATE NOT NULL,
  added_by VARCHAR(100)
);

```

```
-- Table 2: Log every chart view event
CREATE TABLE chart_view_log (
  log_id BIGINT PRIMARY KEY AUTO_INCREMENT,
  chart_id INT NOT NULL,
  user_id VARCHAR(100),
  view_timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
  session_id VARCHAR(100),
  FOREIGN KEY (chart_id) REFERENCES dashboard_charts(chart_id)
);
```

Zombie Chart Detection Query:

```
-- Identify zombie charts: 0 views in last 30 days
SELECT
  dc.chart_id,
  dc.dashboard_name,
  dc.chart_name,
  dc.chart_type,
  dc.added_date,
  COALESCE(MAX(cvl.view_timestamp), 'Never Viewed') AS last_viewed,
  COUNT(cvl.log_id) AS total_views_all_time,
  'ZOMBIE - Recommend Removal' AS action_flag
FROM
  dashboard_charts dc
LEFT JOIN
  chart_view_log cvl
  ON dc.chart_id = cvl.chart_id
  AND cvl.view_timestamp >= DATE_SUB(CURDATE(), INTERVAL 30 DAY)
GROUP BY
  dc.chart_id, dc.dashboard_name, dc.chart_name, dc.chart_type, dc.added_date
HAVING
  COUNT(cvl.log_id) = 0 -- Zero views in last 30 days
ORDER BY
  dc.added_date ASC; -- Oldest zombie charts first

-- LEFT JOIN ensures charts with NO views at all still appear
-- HAVING filters for the 30-day window zero-view condition
-- This output feeds directly into the "Evidence approach" for chart removal
```

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Quick Reference: Topic Summary

Topic	Core Takeaway	Key Term
1.1 KPIs	Use BOTH lead (predictive) and lag (resultant) indicators	Lead vs. Lag
1.2 Five-Second Rule	Critical insight must be grasped in 5 seconds — use layout hierarchy	Information Hierarchy

1.3 Gestalt Principles	Group (Proximity), style (Similarity), section (Enclosure)	Visual Perception
1.4 Color Theory	Color = meaning not decoration. Traffic lights, max 5-7 colors	Categorical/Sequential/Diverging
1.5 Chart Selection	Match chart type to: Comparison, Trend, Composition, Correlation	Chart Category Framework
1.6 User Stories	Design for a specific person, not "everyone". Know WHO before WHAT	As a [user] I want...
1.7 Interactivity	Drill-downs for depth, tooltips for context, page nav for audiences	Drill-Down / Tooltip / Slicer
1.8 Request Bloat	Every chart must pass Decision + Audience + Frequency tests	HiPPO Effect / 3-Filter Test



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